

Omar Bakr

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Education

McMaster University

Hamilton, ON

B.Eng. Software Engineering

Expected April 2027

- **Relevant Coursework:** Computer Architecture, Concurrent Systems, Large System Design, Software Testing

Experience

Manulife

Jan 2026 – Dec 2026

Software Engineer Intern

Toronto, ON

- Built **Python automation and backend services** on **AKS**, reducing manual workflow processing time by **30%**.
- Developed **Terraform automation** to provision and manage SANs, reducing request turnaround from days to **1 hour**.
- **Owned migration of Flux GitOps to Terraform**, standardizing infrastructure across **10+ clusters**.
- Contributed to **Snyk log migration**, improving vulnerability visibility across **20+ services**.
- Designed a **SAST/SCA security scanning framework** integrating **SonarQube** and **Snyk** into **Jenkins CI/CD**, enabling automated vulnerability detection across application repositories.

Definity

May 2025 – Aug 2025

Software Engineer Intern

Waterloo, ON

- Engineered a **Bitbucket automation pipeline** provisioning env variables, cutting setup time by **85%**.
- Built a **Kubernetes CronJob with Bash** to clean up unused ReplicaSets across all namespaces, packaged it as a reusable **Helm chart**, and automated multi-cluster rollout using **ArgoCD**, reducing manual maintenance.
- Built a **Bitbucket and Bash** pipeline to audit and sync **Vault creds**, uncovering **100+ desynced secrets**.

Canada Life

May 2024 - Aug 2024

Software Engineer Intern

Toronto, ON

- Automated DB refreshes with **FastAPI and SQL**, saving **2 hours daily** for five developers on the team.
- Managed **30+ repos**, resolved **150+ critical vulns** with **Veracode**, ensuring compliance with security standards.
- Built **CI/CD pipelines** in GitLab to scan and deploy **50+ microservices** to **EKS**, cutting deployment time by **40%**.

Vidyard

Jan 2024 - Apr 2024

Software Engineer Intern

Waterloo, ON

- Migrated infra from **ECS to EKS** with **Istio, Karpenter, and ArgoCD**, reducing costs by **22%** (\$30K annually).
- Led development of an AI support tool with **Langchain and vector DBs**, integrating Slack, GitHub, and Confluence as knowledge sources to streamline onboarding and improve developer support response by **50%** for **100+ engineers**.
- Built features in **JS and Ruby**, leveraging **CI/CD** with GitHub Actions on **AWS and GCP**.

FirstHX

May 2023 - Dec 2023

Software Engineer Intern

Toronto, ON

- Deployed **Oracle Cloud** infra with **Terraform and GitLab CI**, using **blue-green** to eliminate downtime.
- Built secure endpoints for **2FA and password recovery** with **React, Node.js, TypeScript, and JWT**.
- Integrated **Cypress** for frontend testing, increasing test coverage and reducing bugs by **70%**, streamlining automation.

Projects

Slop Block | Chrome extension to filter low-signal LinkedIn content – **DEMO VIDEO**, **EXTENSION**

- Built a **Chrome MV3 extension** that filters LinkedIn feeds using heuristics and on-device **LLM classification**.
- Designed a **two-stage pipeline**: fast keyword-based filtering with an AI fallback for ambiguous posts.
- Implemented a **non-destructive UI overlay** (blur + reveal) that preserves DOM structure.

MacShuttle | Real-time shuttle ETA app – **PITCH VIDEO (McMaster)**, **DEMO SLIDES (Attridge)**

- Led a team of **5** to build a shuttle ETA system processing **10K+ GIS datapoints** for precise predictions.
- Presented to McMaster administrators; now in **licensing talks** with Attridge, serving **10K+ daily riders**.
- Scaled backend with **Go + PostgreSQL** on **AWS**, automated CI/CD via GitHub Pipelines, and validated **Cypress**.

Technical Skills

Languages: Python, Go, JavaScript/TypeScript, SQL, Bash

Frameworks: React, React Native, FastAPI, Node.js, PostgreSQL, AWS, Kubernetes, Terraform, CI/CD (GitHub/GitLab)